

Soumyaranjan Dash

Knowledge City, Sector-81, Mohali, Punjab, India, 140306
+919078927598 | ph21067@iisermohali.ac.in



OBJECTIVE

To be associated with a progressive organization that allows me to apply my knowledge and skills to keep abreast with the latest trends and technologies.

EDUCATION

Indian Institute of Science Education and Research Mohali

2021-Ongoing

PhD in Condensed Matter Theory

Utkal University, Odisha, India

2018-2020

Masters in Physics

8.54 CGPA

Utkal University, Odisha, India

2015-2018

BSc in Physics

9.40 CGPA

RESEARCH TOPIC

Strongly Correlated and Quantum Many-Body Systems

Exploration of interacting quantum systems where electron correlations lead to emergent collective behavior. Focus on understanding how competing interactions give rise to novel phases of matter beyond conventional band theory. Emphasis on identifying universal mechanisms underlying complex quantum phenomena.

Magnetism and Competing Interactions

Study of magnetic systems with competing exchange interactions that produce rich and nontrivial phase behavior. Investigation of ordering, fluctuations, and field-induced transitions in low-dimensional systems. Aim to understand how microscopic interactions govern macroscopic magnetic properties.

Transport and Response Properties

Analysis of how quantum systems respond to external perturbations such as fields and temperature. Study of transport coefficients and response functions to connect theoretical models with experimentally measurable quantities. Emphasis on uncovering signatures of underlying interactions in observable properties.

Exotic Spin Textures in Metallic Magnets

Study of complex spin configurations emerging in itinerant electron systems due to the interplay of exchange interactions and electronic degrees of freedom. Investigation of nontrivial textures such as noncollinear and topological spin structures stabilized by competing interactions and external fields. Emphasis on understanding their impact on electronic transport and response properties. Exploration of how these textures arise, evolve, and couple to charge dynamics in metallic environments.

PROGRAMMING LANGUAGE

Fortran

C

C++

Python

Mathematica

Julia

OPERATING SYSTEM

Linux
Unix
Windows
MacOS

ACHIEVEMENTS & AWARDS

Best poster award in national conference PSCES'24
Best Poster award in national conference MMT-2019
Joint CSIR-UGC Junior Research Fellowship-2022, Government of India
All India Graduate Aptitude Test for Engineering, with 97.06 percentile
All India Joint Entrance Screening Test with 98.56 percentile

TEACHING EXPERIENCE

Teaching assistant Electronics Lab (Course Code : PHY312)	01/2025 - 05/2025
Teaching assistant Electricity and Magnetism (Course code: PHY102)	01/2024 - 05/2024
Teaching assistant Theory of Mechanics(Course code:PHY101)	07/2023 - 12/2023
Teaching assistant Modern Physics Lab (Course code: PHY212)	01/2023 - 05/2023
Teaching assistant Optics Lab (course code: PHY211)	07/2022 - 12/2022

PUBLICATIONS

Vacancy Induced Expansion of Spin-Liquid Regime in J1–J2 Heisenberg Model
Phys. Rev. Lett. 136, 076502 (2026)

Semiclassical Approach to Partial Magnetic Order in Kondo Lattices
Phys. Rev. B 111, 174443 (2025)

Skyrmions and Magnetic Bubbles in Spin-Orbit Coupled Metallic Magnets
Phys. Rev. B 108, 184419 (2023)

Valence Bond Glass and Glassy Spin Liquid in Disordered Frustrated Magnets
arXiv:2604.05501 (cond-mat)

Thermal Evolution of Skyrmions in Synthetic Ferrimagnets of Co/Gd Heterostructure for Topological Spintronic Applications
arXiv:2512.24091 (cond-mat)

CONFERENCES/WORKSHOPS

National Conference on Physics of Strongly Correlated Electron System
IISER Mohali, Punjab, India, May 13 - May 15, 2026

Advanced School and Conference on Quantum Matter
ICTP, Trieste, Italy, Dec 01 - Dec 12, 2025

Current Trends in Strongly Correlated and Frustrated Systems
MPIPKS, Dresden, Germany, Nov 10 - Nov 14, 2025

Engineered 2D Quantum Materials
ICTS, Bengaluru, India, July 12 - July 26, 2024

National conference on Physics of Strongly Correlated Electron System
IISER Bhopal, India , April 08 - April 10, 2024

School on condensed matter meets quantum information.
ICTS, Bengaluru, India, Sept 25 - Oct 06, 2023

Workshop on computational modeling of Materials

Andhrapradesh, India, May 08 - May 12, 2023

National conference on Trends in Minerals and Matrials Technology

ITER-SOA, Bhubaneswar, India , Oct 30 - Nov 1, 2019

REFERENCE

Sanjeev Kumar - IISER, Mohali

Professor

sanjeev@iisermohali.ac.in

Yogesh Singh - IISER, Mohali

Professor

yogesh@iisermohali.ac.in